

Adaptation Measures and Cost-Benefit Analysis: Brazil Case Study (Porto Alegre)

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Abstract

This report documents the extension of the Brazil case study of the IDB RiskHub / CLIMADA (UNU-EHS) Starter Pack, covering flood risk in Porto Alegre, Rio Grande do Sul, for three asset types: Schools, Residential, and Companies. Building on the existing Schools example, the exercise calibrated Residential and Companies, parametrized six adaptation measures per asset (eighteen in total, covering all six CLIMADA measure mechanisms), and ran a cost-benefit analysis for the period 2025 to 2050. The cost-benefit analysis uses a return-period depth hazard, so Residential and Companies are assigned a depth-damage curve from the literature [2], matched by physical unit to that hazard and consistent with the curve already used for Schools. Because this curve's amplitude is a regional literature default, it is calibrated locally by anchoring it to the real loss observed during the 2024 flood event, using the return-period depth raster whose flooded extent most closely matches that event as the physical reference point. The resulting totals are: total climate risk of 320.348 million USD for Schools, 470.267 million USD for Residential, and 828.925 million USD for Companies. Under this risk estimate, Companies' six measures remain economically justified, while none of Residential's six measures, at their currently parametrized cost, pay for themselves. We trace this asymmetry to two compounding causes: several of Residential's measure costs reflect city-scale programme budgets that are large relative to its real risk, and Residential's risk, once anchored to observed loss, is small relative to its very large total exposed value. This is a decision-relevant finding, not a modelling failure.

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1 Introduction

The IDB RiskHub intersessional assignment asks each participant to work with one or more calibrated asset types from their country’s Starter Pack, select three adaptation measures relevant to the case study (Brazil participants: three *additional* measures, since Schools already shipped with three), parametrize those measures in the Entity file with realistic implementation and maintenance costs and a defined effect on the risk calculation, run a cost-benefit analysis using the Brazil example notebook, refine the parametrization until the results appear realistic, and prepare figures suitable for presentation to decision makers. Participants are further encouraged to calibrate any remaining uncalibrated assets in their case study, and to note ideas for improving the Starter Packs.

For Brazil, the three-additional-measures rule applies because the Schools entity already shipped with three measures (river dredging, levee rebuilding, and an early warning system). Residential and Companies had no real pre-existing measures, only an inert placeholder row, so the rule was applied uniformly across all three asset types: three baseline measures plus three additional measures, six total per asset, eighteen in total. Applying it uniformly rather than adding three measures only to Schools keeps the three assets directly comparable, which matters because the most decision-relevant results in this report come from contrasting how the same six mechanisms perform against three very different exposure bases.

This report covers four areas of work: the calibration of Residential and Companies, which was not previously available in the pack; the parametrization of six adaptation measures per asset, covering all six CLIMADA measure mechanisms; the derivation and local calibration of the depth-damage curve used for Residential and Companies in the cost-benefit analysis; and the production of figures and tables suitable for decision makers. A companion investigation, prompted by the result that none of Residential’s measures pay for themselves, confirmed that this is not a reproducibility issue (Section 4 and the verification notes in Appendix A).

2 Methodology

2.1 Environment

The analysis uses CLIMADA version 6.1.0 [1], pinned deliberately: version 7, expected later in 2026, rewrites the Measures and CostBenefit modules used throughout this exercise. Anyone reproducing these results should pin the same version rather than take the latest release, since the Entity-file and cost-benefit interfaces this exercise depends on are the ones scheduled to change. The commands below create that environment from scratch and end with the library’s own test suite, which should report OK before any notebook is run.

```
conda create -n climada_env --override-channels -c conda-forge climada=6.1.0 \
    notebook nbconvert -y
conda activate climada_env
python -m unittest climada.engine.test.test_impact    # must end in "OK"
```

2.2 Baseline Reproduction

The pack’s original notebook, `09_costbenefit.ipynb`, was executed unmodified to confirm the environment. It reproduced the documented reference numbers exactly: total climate risk 320.348 million USD, average annual risk 19.0853 million USD, residual risk minus 165.334 million USD, using three measures (river dredging, levee rebuilding, early warning system) on the Schools asset. As an independent check on this session’s own environment, the same notebook was re-executed a second time, from a pristine clone of the upstream Starter Pack repository, with zero modifications from any working session. It reproduced the same three figures exactly,

confirming that the CLIMADA library and the cost-benefit method work correctly and are not sensitive to local environment peculiarities.

That same check on the upstream repository showed that the Starter Pack, as published, ships a complete cost-benefit example only for Schools. Residential and Companies exist upstream only as uncalibrated Entity files, with no measures and no prior cost-benefit analysis of any kind. The calibration and cost-benefit work for those two assets, described in the remainder of this report, is therefore a first attempt, not a reproduction of an existing, previously validated result.

2.3 Calibration of Residential and Companies

Schools was already calibrated in the pack; Residential and Companies were not. The method follows the one used in `06a_calibration_schools.ipynb`: a two-dimensional grid search over the parameters `max_damage` and `max_intensity`, scaling a first-guess curve against the 2024 event’s flood-duration hazard, followed by a fine one-dimensional refinement once the coarse grid identifies the neighborhood of the optimum. The search space is set independently for each asset rather than reused from Schools, because the three assets differ enormously in exposure scale and in the ratio between exposed value and observed loss. Residential, for example, is a whole-city grid, with 69.3 billion USD of exposed value against 212 million USD of observed loss, about 0.3 percent, so the search space is widened to allow `max_damage` down to 0.01, well below the range that would be appropriate for an asset with a higher damage ratio such as Schools. Companies, with a smaller and more concentrated exposure base, is calibrated over its own search space in the same way. The resulting fits, summarized in Table 1, reproduce the observed 2024 loss for both assets to within one percent.

Table 1: Calibration results for Residential and Companies. Only `max_damage` is identified by the data; `max_intensity` is not, for the reasons given below.

Asset	<code>max_damage</code> (identified)	Modelled loss (USD)	Observed loss (USD)	Error
Residential	0.028	212,265,415	212,170,680	0.04%
Companies	0.296	442,034,021	438,755,400	0.75%

Table 1 deliberately reports only `max_damage`, even though the grid search fits two parameters. The reason is that only `max_damage` is actually identified by the data. This is a known property of the method, documented in the pack’s own 06a notebook: calibrating two free parameters against a single aggregate target leaves one degree of freedom, so for every `max_intensity` there exists a `max_damage` that reproduces the observation. Our own sweep confirms it directly: holding `max_intensity` fixed at values from 0.25 to 79.75 days, spanning the entire searched range, a fine search over `max_damage` still reaches an error under 3 percent for Residential and under 0.5 percent for Companies at every value tested.

The mechanism behind this degeneracy is specific to the severity of the 2024 event. Median flood duration at the exposure points is 48 days for Residential and 29 days for Companies, while the scaled curve reaches its maximum damage at roughly `max_intensity` divided by three. Consequently the great majority of flooded exposed value, about 93 percent for Residential and 91 percent for Companies at the values the search selected, sits on the flat, saturated part of the curve, where modelled loss no longer depends on duration at all. The model effectively collapses to the product of `max_damage` and the total flooded exposed value, which pins `max_damage` analytically at 0.0277 for Residential and 0.2942 for Companies, within 1.2 and 0.4 percent of the values in Table 1, and leaves `max_intensity` almost unconstrained.

Two diagnostics confirm that the fitted `max_intensity` carries no information. Both assets settled on opposite edges of the permitted box, Residential at 28.75 days, the largest value the

inherited bound allows, and Companies at 0.25 days, the first point of the grid: the signature of an optimizer sliding along a flat objective until it meets a constraint. The Companies value is also physically implausible on its own terms, since a curve saturating at 0.25 days would imply that a business flooded for two hours suffers the same damage as one flooded for the full 91 days the event lasted. The stored curves retain these values (28.75 and 0.25 days) for traceability with the notebooks, but they are reported here as an artifact of the search, not as an estimate. Resolving the ambiguity properly would require the same kind of asset-specific expert judgement the pack applies to Schools, which in turn requires local evidence on how residential and commercial losses scale with flood duration that this exercise does not have. None of this affects the cost-benefit results in Section 3, since the day-based curve is not the one used there.

These calibrated curves, stored under `data/entity_files_calibrated/`, are valid for reconstructing the single 2024 event, whose hazard footprint is measured in flood duration (days). The cost-benefit analysis, by contrast, runs on a return-period depth hazard (meters); it therefore uses a separate, depth-based damage curve, described in Section 2.5, matched by physical unit to that hazard. Keeping the two curves in separate files rather than overwriting one with the other is deliberate: each remains correct for its own hazard, and the separation makes it impossible to feed a duration-calibrated curve into a depth-based calculation by accident.

2.4 Adaptation Measures and Mechanisms

Each asset received six measures: three that shipped with the pack or follow the same mechanism family for Residential and Companies, and three additional measures selected for this exercise, following the intersessional brief’s instruction to select measures relevant to the case study and parametrize their effect on the risk calculation explicitly. Rather than varying only the numeric parameters of a single mechanism, the eighteen measures were designed to cover all six adaptation mechanisms CLIMADA supports, spread across the three assets so that each mechanism is demonstrated at least once. This choice is deliberate: a decision maker comparing measures that differ only in cost learns less than one comparing measures that intervene on the risk calculation in genuinely different ways.

Table 2: CLIMADA measure mechanisms demonstrated in this exercise

Mechanism	Example used
Linear hazard transform (a, b)	School floor raising, home flood-proofing, portable barriers
Hazard frequency cutoff (RP)	Rebuilding levees, city-wide drainage upgrade, commercial levee
Risk transfer (insurance)	Portfolio, residential parametric, and business interruption insurance
Exposures swap (relocation)	Partial relocation in all three assets
Linear MDD transform	Early warning system, community warning, elevated inventory storage
Linear PAA transform	Flood-resistant building codes, business continuity planning

For insurance, attachment and cover limits were derived from the model’s own exceedance curve (`Impact.calc_freq_curve`, using the 10-year return period as attachment and the 100-year return period as cover), not from arbitrary values. The cost loading applied was 1.5 times the pure premium, which is why the insurance benefit-to-cost ratio sits close to 1 divided by 1.5, approximately 0.66, across all three assets. This value reflects the cost of transferring volatility,

not an expectation of savings.

For relocation, an HDF5 Exposures file was built in which the locations with the highest depth at the 100-year return period have their value reduced to 40 percent of the original, corresponding to a partial relocation of 60 percent of the exposed value. A second variant was built for 2050, already scaled by the 2 percent annual economic growth rate; without this adjustment, the measure’s benefit would incorrectly include the removal of economic growth itself from the future scenario.

2.5 Depth-Damage Curve for the Cost-Benefit Analysis

Residential and Companies are assigned a depth-based damage curve for the cost-benefit analysis, matched to the return-period depth hazard used there. Its shape comes from the curve published by the Joint Research Centre [2]: Table 3-5, for residential use in South America, is adopted for Residential, and Table 3-9, for commercial use in South America, is adopted for Companies. This is the same literature source and the same pattern already used, though without explicit citation, by the Schools entity shipped with the pack, which uses an equivalent depth-based curve.

Because a regional literature curve’s amplitude is not, by itself, validated for Porto Alegre, it is calibrated locally against the only available local data point: the real loss observed during the 2024 flood event. No depth-based footprint exists for that event, only a duration-based one, so a physical proxy is needed. Among the available return-period depth rasters, the one whose flooded extent most closely matches the extent actually observed in 2024 serves as that proxy. Comparing flooded area across rasters (pixel count with depth greater than zero, converted to square kilometers accounting for Porto Alegre’s latitude), the 2024 event’s extent (141.2 km²) falls almost exactly between the 100-year return period (134.3 km²) and the 500-year return period (148.0 km²), closest to the 200-year return period (140.9 km², a difference of only 0.3 km²). This is consistent with public accounts of the 2024 Rio Grande do Sul floods as an exceptional, near record-breaking event for Porto Alegre, and the 200-year raster is used as the anchor.

The curve’s amplitude is calibrated by rescaling: keeping the same depth breakpoints and the same relative shape of the JRC curve, every mean damage degree (MDD) value is multiplied by a single scale factor, computed as the ratio of the observed 2024 loss to the loss the unscaled curve would produce on the 200-year return-period raster (0.048339 for Residential, 0.360351 for Companies). Because no value of the rescaled curve saturates at MDD equal to one within the observed depth range (0 to about 9 meters), this scaling propagates exactly linearly through the rest of the analysis: total risk, average annual risk, and each measure’s benefit (with the exception of insurance) are each the value the unscaled curve would have produced, multiplied by the same asset-specific scale factor. Insurance attachment and cover limits are computed from this calibrated curve’s own exceedance curve, following the same method as Section 2.4.

The resulting totals are 470.267 million USD for Residential and 828.925 million USD for Companies (Section 3). Measure costs are parametrized independently, as construction or municipal programme budgets, not as a quantity derived from the risk scale; Section 4 examines the consequence of that independence in detail: none of Residential’s six measures pay for themselves at their parametrized cost, while all six of Companies’ measures continue to. This is an explicit limitation to keep in mind: the curve is anchored to a single aggregate data point (one event’s total loss), not to a spatially distributed calibration like the one performed for the duration hazard in Section 2.3, the same caveat that applies to any calibration built on few reference points.

As an independent check that this scale factor is not an arbitrary or forced choice, the same calibration was reproduced using CLIMADA’s own calibration framework (`climada.util.calibrate`), which minimizes the mean squared error between modelled and observed loss over a free scale parameter using `scipy.optimize`. Three optimization algorithms (Nelder-Mead, L-BFGS-B,

and Powell) were run independently and all converged to the same values obtained analytically, 0.048339 for Residential and 0.360351 for Companies, confirming that these are the cost-minimizing values for this one-parameter model given the observed 2024 loss, not a value chosen to produce a particular outcome.

2.6 Sensitivity to the Anchor Return Period

The scale factor is sensitive to which return period is used as the physical proxy for the 2024 event: computed the same way at every available return period, it ranges from 2.37 (Residential, RP10) down to 0.039 (Residential, RP500), since the raw curve’s modelled loss grows with depth while the observed 2024 loss is fixed. This makes the RP200 choice, based on flooded-area similarity (Section 2.5), the single most consequential methodological decision in this analysis, and it is worth checking how much the reported results would change under a different, still physically plausible choice.

Table 3: Total risk and non-insurance measure B/C under alternative anchor return periods

	RP100	RP200 (used)	RP500
Residential total risk (M USD)	569.0	470.3	378.1
Residential measures, B/C range	0.23 to 0.88	0.19 to 0.73	0.15 to 0.59
Companies total risk (M USD)	1,019.8	828.9	632.5
Companies measures, B/C range (excl. relocation)	4.37 to 11.17	3.55 to 9.08	2.71 to 6.93
Companies relocation, B/C	1.62	1.32	1.01

Across this entire range, spanning a 100-year to 500-year event, the decision-relevant conclusion is stable: every one of Residential’s non-insurance measures stays below a benefit-to-cost ratio of 1, and every one of Companies’ measures other than relocation stays above 1. Companies’ relocation measure is the one result sensitive enough to flag on its own: its ratio moves from 1.62 at RP100 to 1.01 at RP500, staying on the right side of break-even across the tested range but by a narrow and anchor-dependent margin. Insurance’s ratio is excluded from this table because it depends on the model’s exceedance curve rather than scaling linearly with the anchor choice; in practice it stays close to 0.66 under every return period tested, since it is set by the 1.5 times cost loading rather than by the underlying risk level.

2.7 Acceptance Criteria

Each measure’s parametrization is checked against three criteria: no infinite, undefined, or greater-than-50 benefit-to-cost ratio, which would signal an unrealistically low cost; no individual measure’s benefit exceeding the asset’s total climate risk; and insurance attachment and cover limits always derived from the model, never from arbitrary estimates. The first two are sanity checks against a mis-specified cost or a benefit computed against the wrong baseline; the third exists because risk-transfer parameters chosen by hand can produce almost any premium, and with it almost any benefit-to-cost ratio, without the error being visible anywhere in the output.

Under the final parametrization, no benefit-to-cost ratio exceeds 9.08, well below the ceiling of 50, and no individual benefit exceeds the asset’s total climate risk; both sanity criteria are satisfied. For Residential, all six ratios fall below 1.0, which is not a sign of a parametrization error: it is the honest conclusion that, at this risk estimate, these specific measures, at the costs they were parametrized with, do not pay for themselves within the study horizon. Adjusting the costs to force ratios above one would substitute the desired outcome for the best available estimate of cost and risk, which is exactly what these criteria are meant to prevent.

3 Results

Each of the three subsections below covers one asset type in turn: Schools, Residential, and Companies. Every subsection follows the same structure, so results across assets can be compared directly. It opens with the asset’s total climate risk and average annual risk for 2050, then a table listing all six adaptation measures with their mechanism, cost, benefit, and benefit-to-cost ratio, then the five figures generated for that asset (the cost-benefit chart, benefits by return period, the waterfall of averted risk, the expected annual impact map, and the depth map with relocation points), and closes with a paragraph interpreting what those figures show. A final subsection, Cross-Asset Synthesis, draws out the patterns that only become visible when comparing all three assets side by side, such as the stability of the insurance ratio and the climate-versus-growth split.

3.1 Schools

Schools is the one asset in this exercise whose calibration and damage curve were never touched: its risk figures below are exactly those the Starter Pack shipped with, unaffected by the calibration and curve-derivation work described in Section 2 for Residential and Companies. Total climate risk is 320.348 million USD, with an average annual risk of 19.0853 million USD; both numbers are reproduced exactly by the pristine, unmodified upstream notebook, as described in Section 2.2. Table 4 lists all six measures, three that shipped with the pack (river dredging, levee rebuilding, and the early warning system) and three added for this exercise (school floor raising, portfolio insurance, and partial relocation), spanning five of the six CLIMADA measure mechanisms on their own.

Table 4: Schools: adaptation measures and cost-benefit results

Measure	Mechanism	Cost (M USD)	Benefit (M USD)	B/C
River dredging	hazard -1.5 m	789.562	195.782	0.25
Rebuilding levees	RP cutoff ≈ 100 y	161.171	257.866	1.60
Early warning system	MDD $\times 0.9$	1.500	32.035	21.36
School floor raising	hazard -0.4 m	7.625	64.244	8.43
Portfolio flood insurance	risk transfer 32.3M xs 116.9M	258.463	171.475	0.66
Partial relocation (5 of 122)	exposures swap	10.500	34.764	3.31

Costs in Table 4 range from 1.5 million USD for the early warning system to 789.6 million for river dredging, and benefit-to-cost ratios range even more widely, from 0.25 for dredging up to 21.36 for the early warning system, the two ends of a pattern that recurs across all three assets: hazard-transform measures with no protection ceiling are expensive relative to the risk they avert, while cheap, narrowly targeted measures post the highest ratios precisely because their cost base is so small. The five figures that follow show this pattern from different angles, from the raw cost-benefit comparison to how each measure’s contribution changes with event severity.

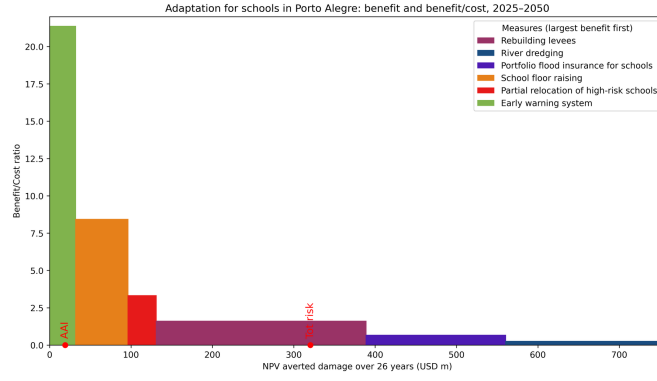


Figure 1: Cost-benefit results, Schools

Figure 1 plots each measure as a rectangle whose width is the net present value of averted damage and whose height is the benefit-to-cost ratio, so area is proportional to neither quantity alone and the two dimensions must be read together. The early warning system is the tall, narrow bar on the left: an excellent ratio built on a very small absolute benefit. River dredging is the opposite, the widest bar in the chart but so flat that it sits below the break-even line. For a decision maker this is the central trade-off in the Schools portfolio, since the measure that averts the most damage in absolute terms is also the one that fails to justify its cost.

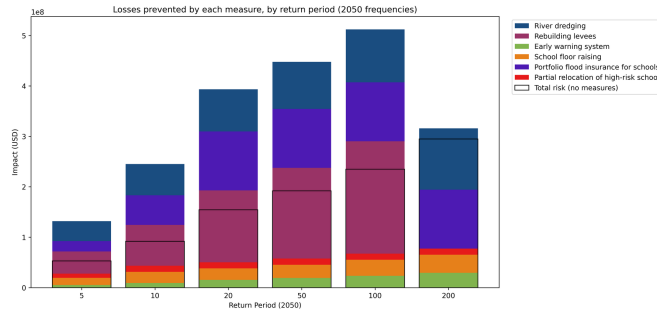


Figure 2: Benefits by return period, Schools

Figure 2 breaks the same benefits down by event severity, and explains why dredging looks so different from the rest. As a hazard transform with no protection ceiling, it keeps growing in absolute terms as events get more severe, and it dominates the benefit at the 200-year return period. Levees, with a cutoff near the 100-year return period, and insurance between them absorb most of the average risk across the middle of the range. Relocation and the early warning system appear only as thin bands throughout, which is consistent with their small absolute benefits even where their ratios are favourable.

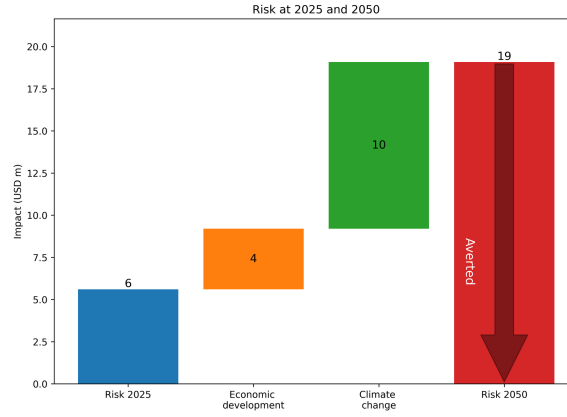


Figure 3: Waterfall of averted risk, Schools

Figure 3 separates the drivers of future risk rather than the effect of the measures. Of the 19 million USD of risk projected for 2050, 6 million already existed in 2025, a further 4 million comes from economic growth, and a further 10 million comes from climate change. Roughly 71 percent of the growth in risk between now and 2050 is therefore attributable to the climate scenario rather than to more assets being built in exposed places, a proportion that recurs closely in the other two assets and is discussed in Section 3.4.

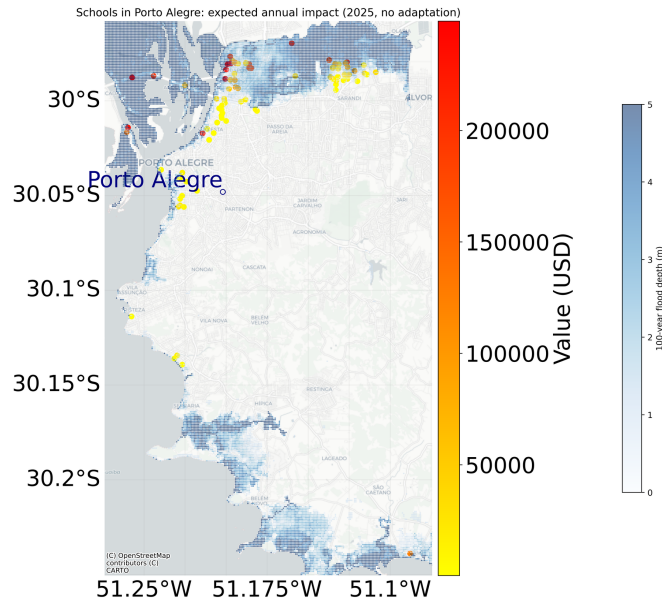


Figure 4: Expected annual impact map, Schools

Figure 4 moves from aggregate to spatial, showing where the expected annual impact actually falls. Risk concentrates in the Farrapos and Humaita neighborhoods and around Sarandi, the lowest-lying fringes along Lake Guaíba, with the rest of the school network contributing very little. Because Schools is a point-facility dataset of 122 locations rather than a grid, individual high-risk schools are visible as discrete markers, which is what makes a targeted measure such as relocation feasible to specify in the first place.

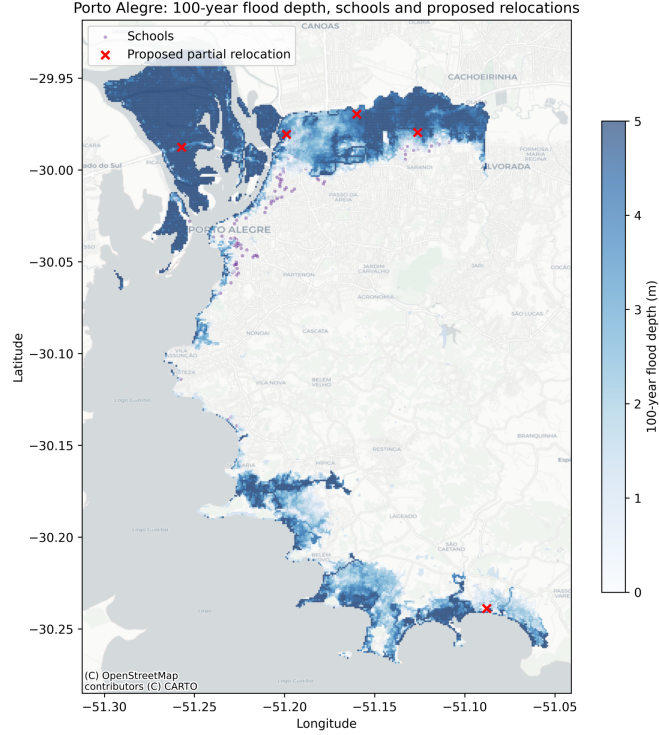


Figure 5: 100-year return period depth and relocation points, Schools

Figure 5 overlays the 100-year flood depth on the school locations and marks the five schools selected for partial relocation. They sit exactly where the depth footprint is darkest, which is the criterion by which they were chosen, with one isolated case further south along the lake. The clustering explains why relocating only five of 122 schools still returns a benefit-to-cost ratio above three: risk in this asset is concentrated enough that a small, well-targeted intervention captures a disproportionate share of it.

3.2 Residential

Residential is the asset most affected by the depth-damage curve derivation in Section 2.5, since it is calibrated and risk-assessed for the first time in this exercise rather than reusing an existing pack example. Total climate risk is 470.267 million USD, with an average annual risk in 2050 of 28.0255 million USD, figures already anchored to Porto Alegre's real 2024 loss and checked for sensitivity to the anchor choice in Section 2.6. Table 5 lists the six measures parametrized for this asset: home flood-proofing and a city-wide drainage upgrade, both carried over in mechanism from the Schools example; a parametric insurance layer sized to this asset's own exceedance curve; partial relocation of the highest-risk households; and two measures added specifically for Residential, a community-based early warning system and flood-resistant building codes for renovations.

Table 5: Residential: adaptation measures and cost-benefit results

Measure	Mechanism	Cost (M USD)	Benefit (M USD)	B/C
Home flood-proofing	hazard -0.3 m	362.064	70.321	0.19
Residential parametric insurance	risk transfer 47.8M xs 175.4M	382.446	251.631	0.66
City-wide drainage upgrade	RP cutoff = 20y	300.000	218.109	0.73
Relocation (50 of 20,187)	exposures swap	8.250	3.924	0.48
Community-based early warning	MDD $\times 0.85$	150.000	70.540	0.47
Flood-resistant building codes	PAA $\times 0.8$	200.000	94.053	0.47

None of these six measures pays for itself at its parametrized cost, at this risk estimate. This is the central decision-relevant finding of this report, and it is examined in detail in Section 4, which traces it to two compounding causes: the scale of several of these measures' budgets relative to Residential's real risk, and the underlying gap between how much value is exposed and how much of that value is actually at risk once the damage curve is anchored to a real observed loss rather than left at its literature amplitude. The five figures below show the same conclusion from different angles, from the raw cost-benefit comparison to how the underlying risk itself is distributed across the city.

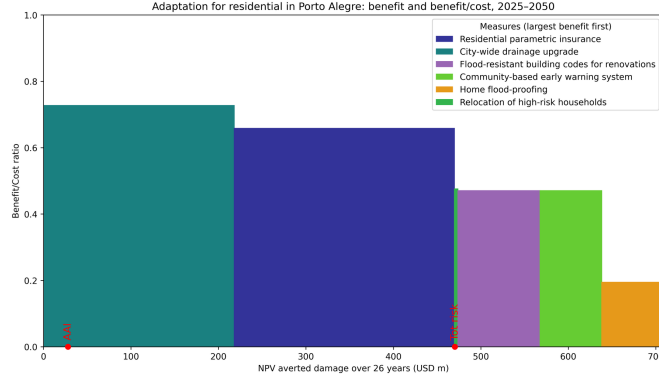


Figure 6: Cost-benefit results, Residential

Figure 6 makes the headline result immediately visible: every rectangle sits below the break-even line, so no measure in this portfolio returns more than it costs. The chart is worth reading alongside the Schools equivalent (Figure 1), where the bars span a wide range of heights. Here they are compressed into a narrow band between 0.19 and 0.73, which is the visual signature of measure budgets that are large relative to the risk they address rather than of any single mis-specified measure.

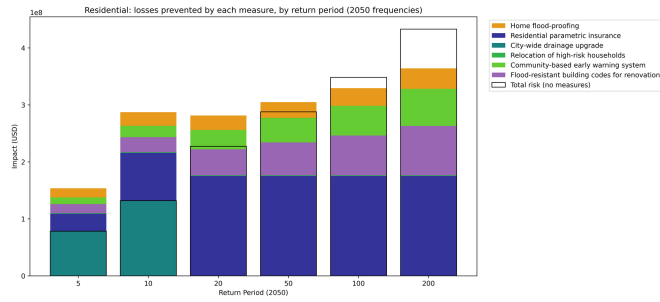


Figure 7: Benefits by return period, Residential

Figure 7 shows a sharp cutoff pattern. The city-wide drainage upgrade, with its cutoff at the 20-year return period, covers the entirety of risk at the 5 and 10-year return periods, below its protection ceiling, then disappears completely from the 20-year return period onward once events exceed that ceiling. Beyond that point insurance, sitting at the highest risk layer, and the building codes take over. At the 200-year return period a slice remains uncovered between the top of the stacked measures and the total risk envelope, showing that the six measures together still leave a visible residual risk against the most extreme event.

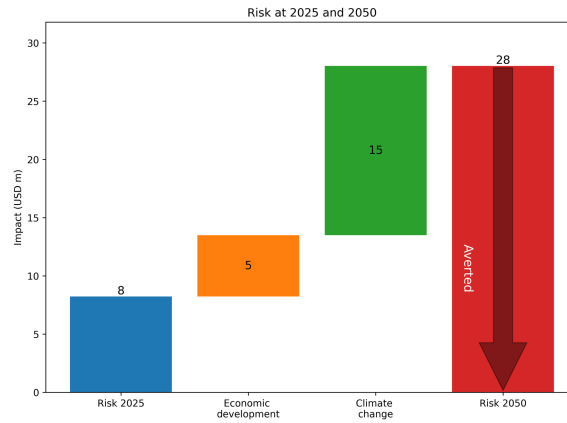


Figure 8: Waterfall of averted risk, Residential

Figure 8 decomposes the growth in risk to 2050. Risk rises from 8.2 million USD in 2025 to 28.0 million in 2050, an increase of 5.3 million attributable to economic growth and 14.5 million to climate change, about 73 percent climate against 27 percent growth. That this matches the Schools split so closely, despite a completely different exposure base and calibration, is one of the more robust findings of the exercise and is taken up in Section 3.4.

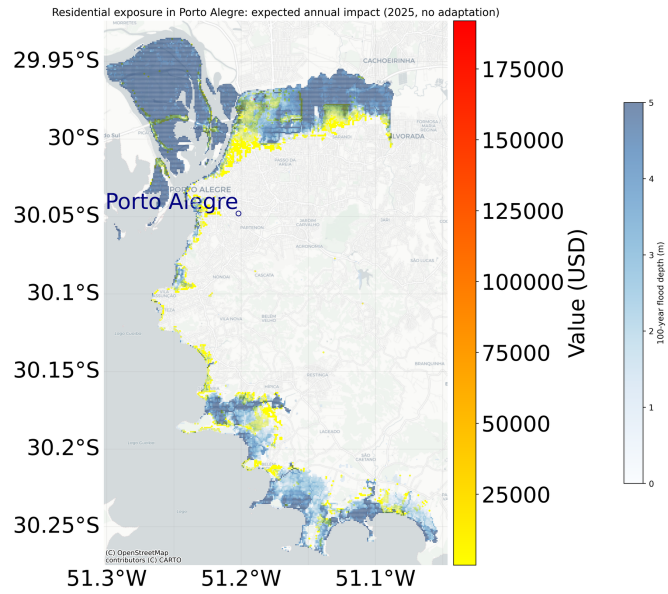


Figure 9: Expected annual impact map, Residential

Figure 9 shows a spatial pattern quite distinct from Schools. Instead of a handful of concentrated points, residential exposure covers nearly the entire urban footprint of Porto Alegre, because this asset is a city-wide gridded dataset of more than 20,000 points rather than a list of facilities. This breadth is precisely what makes city-scale measures expensive here: an intervention defined across the whole grid has to be funded across the whole grid, while the risk it addresses remains concentrated in the low-lying north.

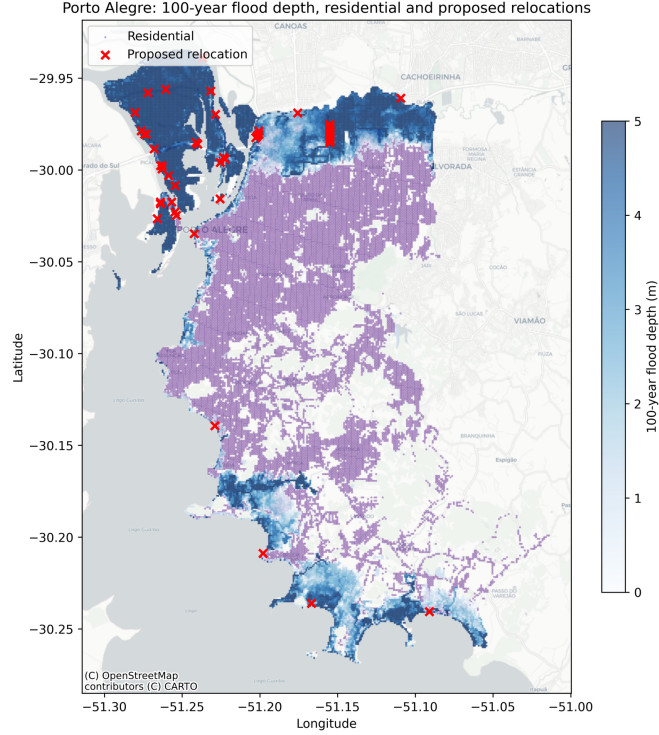


Figure 10: 100-year return period depth and relocation points, Residential

Figure 10 marks the 50 grid points selected for partial relocation against the 100-year depth footprint. Unlike Schools, where the selected sites form a tight cluster, these spread across several riverside neighborhoods with additional pockets further south. The wider spread is one reason relocation performs worse here than for the other two assets: the same conservative rule, moving 60 percent of value at the deepest points, captures a smaller share of total risk when that risk is diffuse rather than concentrated.

3.3 Companies

Companies shares its calibration method and depth-damage curve derivation with Residential, but differs sharply in how concentrated its exposure is: a smaller, denser commercial base along the riverfront and in the historic center, rather than a grid spanning the whole city. Total climate risk is 828.925 million USD, with an average annual risk in 2050 of 49.6197 million USD, both already anchored to the observed 2024 loss in the same way as Residential. Table 6 lists the six measures parametrized for this asset: portable flood barriers and a levee protecting the commercial district, mirroring Residential's hazard-transform and cutoff mechanisms; business interruption insurance; relocation of the highest-risk businesses; elevated inventory storage; and business continuity and supply chain diversification planning.

Table 6: Companies: adaptation measures and cost-benefit results

Measure	Mechanism	Cost (M USD)	Benefit (M USD)	B/C
Portable flood barriers	hazard -0.3 m	36.533	147.182	4.03
Business interruption insurance	risk transfer 66.7M xs 356.6M	713.520	474.347	0.66
Relocation (20 of 4,366)	exposures swap	10.250	13.508	1.32
Levee (commercial district)	RP cutoff = 20y	50.000	321.952	6.44
Elevated inventory storage	MDD $\times 0.8$	18.266	165.785	9.08
Business continuity and supply chain diversification	PAA $\times 0.7$	70.000	248.677	3.55

Unlike Residential, all six measures pay for themselves ($B/C > 1$), a direct consequence of Companies' much smaller, more concentrated exposure base relative to its risk (Section 4 quantifies this contrast directly). Relocation, at 1.32, does so with the narrowest margin of the six, and is the one result in this report sensitive enough to the anchor return period (Section 2.6) to be worth flagging as less certain than the rest; the other five measures stay comfortably above break-even across the full range of anchor choices tested. The five figures below show this pattern in more detail, from the raw cost-benefit comparison through to how the risk itself is distributed across the city.

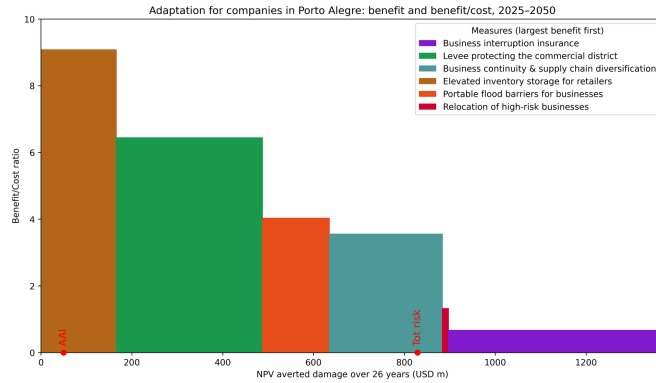


Figure 11: Cost-benefit results, Companies

Figure 11 is the visual counterpart to Figure 6 and the contrast is stark: here the rectangles rise well above the break-even line, with elevated inventory storage reaching 9.08 and the commercial-district levee 6.44. The same six mechanisms, applied to a much smaller and more concentrated exposure base, produce a portfolio that pays for itself several times over. Only insurance sits below the line, and for the structural reason discussed in Section 3.4 rather than because of how it was costed.

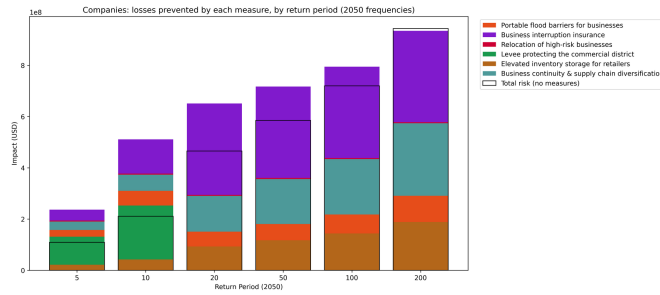


Figure 12: Benefits by return period, Companies

Figure 12 repeats the cutoff pattern seen for Residential. The levee, with a cutoff at the 20-year return period, dominates the benefit at the 5 and 10-year return periods and drops to zero from 20 years onward. Beyond that point insurance dominates at the highest risk layer, while the MDD and PAA transform mechanisms, elevated inventory storage and business continuity planning, contribute a relatively stable middle band across every return period, since neither has a protection ceiling that the event can exceed.

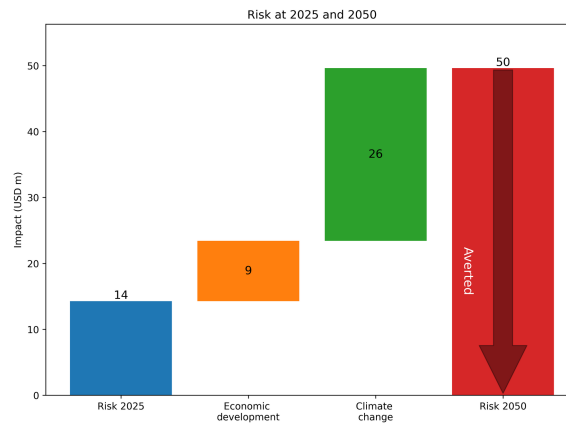


Figure 13: Waterfall of averted risk, Companies

Figure 13 completes the set of three risk decompositions. Risk rises from 14.3 million USD in 2025 to 49.6 million in 2050, with 9.1 million attributable to economic growth and 26.2 million to climate change, close to 74 percent climate against 26 percent growth. This is the third asset to land in the same narrow band despite entirely independent exposure data, calibration, and measures, which is the strongest internal consistency check available in this exercise.

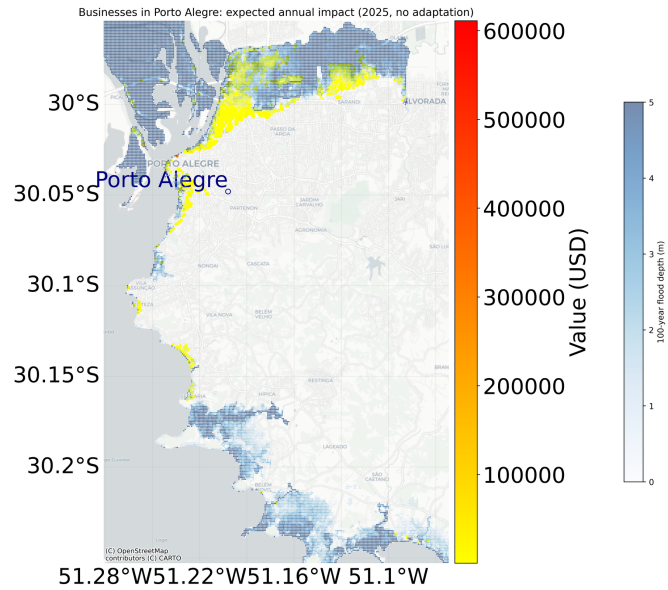


Figure 14: Expected annual impact map, Companies

Figure 14 shows a double concentration rather than the single cluster of Schools or the diffuse spread of Residential. A dense band runs along the river through the Farrapos, Humaita, and Sarandi neighborhoods, coinciding with the lowest and most flood-prone ground, and a second, much smaller cluster sits in Porto Alegre’s historic center, its commercial and port district. That commercial activity concentrates in exactly the places most exposed to flooding is what gives this asset its high risk-to-exposure ratio, and in turn why targeted measures perform so well against it.

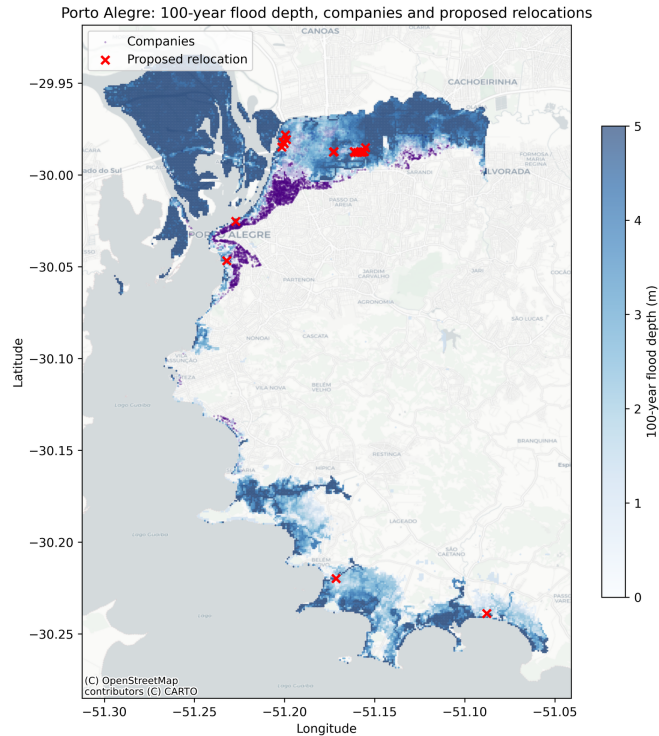


Figure 15: 100-year return period depth and relocation points, Companies

Figure 15 marks the 20 businesses selected for relocation against the 100-year depth footprint. Almost all fall inside the dense riverside band, with only a few isolated points outside it, an intermediate case between the tight clustering of Schools and the wide dispersion of Residential. This is consistent with relocation returning a positive but narrow benefit-to-cost ratio of 1.32 here, the one Companies result flagged in Section 2.6 as sensitive to the anchor choice.

3.4 Cross-Asset Synthesis

Insurance stays consistently close to a benefit-to-cost ratio of 0.66 across all three assets. This is a mechanical consequence of the uniformly applied 1.5 times cost loading, not a modelling failure: insurance's value lies in transferring volatility, not in expected long-run savings, and this ratio depends only on the loading factor, not on the risk scale. It is worth stating explicitly to a decision maker, because a bar chart that places insurance below every other measure invites the reading that it was parametrized badly, when in fact a ratio near the reciprocal of the loading is exactly what a correctly specified risk-transfer layer should produce.

Wide-infrastructure measures, such as levees, dredging, and drainage, show very different benefit-to-cost ratios depending on the asset evaluated. In Schools, these measures protect the entire city but are evaluated only against the narrow slice of school assets, yielding ratios between 0.25 and 1.60. In Companies, the commercial-district levee, evaluated against the population it actually protects, reaches 6.44. In Residential, the same "broader base" logic applies to the drainage upgrade, but its risk scale leaves no margin: its ratio falls to 0.73. In other words, evaluating against a broader base continues to raise the benefit-to-cost ratio relative to Schools, but is not, by itself, enough to offset a real risk much smaller than the exposed value it is measured against.

Relocation is not uniformly positive across the three assets: in Schools the ratio is 3.31; in Companies it is 1.32, positive but by a narrow margin; in Residential it is 0.48, not paying for itself. The conservative approach adopted, relocating only 60 percent of the exposed value at the highest-depth points, is still not enough to justify Residential's relocation cost against

its estimated risk. The pattern across the three assets tracks how spatially concentrated each one’s risk is, as the depth maps in Section 3 show: tightly clustered for Schools, intermediate for Companies, and diffuse for Residential. Relocation rewards concentration, because a fixed number of relocated points captures a larger share of total risk when that risk is not spread thinly.

The proportion between climate change and economic growth in future risk growth is notably stable across all three assets, between 71 and 75 percent attributable to climate change and between 25 and 29 percent to economic growth. This is a sign of the exercise’s robustness: the three assets use completely different exposure bases, calibrations, and measures, but share the same future climate scenario, the 2025 to 2050 return-period remapping, and the same 2 percent annual economic growth rate. The stability of this proportion suggests that the climate scenario, not particular parametrization choices, is the main driver of projected future risk growth.

Finally, measures with a return-period cutoff, such as drainage and levees, always drop to zero above their protection ceiling. This pattern appears clearly in the benefits-by-return-period charts for Residential and Companies: full protection up to the 20-year return period, zero contribution afterward. This is a known limitation of that mechanism, offering complete protection or none, with no partial gradation, and it is worth explaining clearly to a decision maker before the measure is adopted in isolation.

4 Root Cause Analysis: Why Residential Measures Do Not Pay Off

The result that none of Residential’s six measures pay for themselves, while all six of Companies’ do, prompted a dedicated investigation, since a result that surprising warrants verification before it is trusted. Two checks were performed. First, a pristine clone of the upstream Starter Pack repository confirmed that no official Residential or Companies cost-benefit example exists anywhere upstream: only Schools ships a complete, calibrated example. There is therefore no previously ”working” Residential result that this exercise could have regressed; this is the first time this calculation has ever been performed for that asset. Second, the notebook re-execution described in Section 2.2 confirmed the pipeline itself reproduces its own numbers exactly, ruling out an execution or environment bug.

With reproducibility and baseline validity confirmed, the asymmetry itself still needs an explanation, since it is large enough that a reader could reasonably suspect a modelling error rather than a genuine finding. The analysis below identifies two compounding causes, one about how the six measures were costed and one about the underlying economics of the two exposure bases, and quantifies each rather than leaving the explanation qualitative. Table 7 sets out the comparison that motivates both: Residential and Companies differ enormously in exposed value, but far less in total climate risk, and that gap alone accounts for most of what follows.

Table 7: Cost-to-risk comparison, Residential versus Companies

Asset	Exposed value (2025)		Total climate risk		Risk / exposed value	Sum of 6 measure costs		Cost / risk
Residential	69.312	bn USD	470.267	M	0.68%	1,402.760	M	298%
Companies	3.656	bn USD	828.925	M	22.67%	898.569	M	108%

Cause 1: several of Residential’s measure costs are large, programme-scale budgets relative to its risk. Three of Residential’s six measures, Home flood-proofing (362.064 million USD), Community-based early warning (150.000 million USD), and Flood-resistant

building codes (200.000 million USD), are costed as city-scale programmes intended to reach the full residential base, 69.3 billion USD of exposed value spread across more than 20,000 grid points. Only one of Companies’ six measures is costed at a comparable programme scale. This asymmetry in cost scale, independent of any curve calibration question, already accounts for much of the difference in benefit-to-cost outcome between the two assets.

Cause 2: the scale mismatch is also structural. Residential has about 19 times more exposed value than Companies, but its total climate risk is only marginally higher in absolute dollar terms. This makes Residential’s risk intensity, risk as a share of exposed value, about 33 times smaller than Companies’ (0.68 percent against 22.67 percent). The sum of Residential’s six measure costs, 1.4 billion USD, almost three times its own total risk, simply does not fit inside the risk it addresses, even where no individual cost is unusually large in absolute terms.

The difference in how much the unscaled literature curve overestimated the observed 2024 loss, roughly 20.7 times for Residential against 2.8 times for Companies (Section 2.5), traces back to the JRC curve itself: the ”South America residential” table fit Porto Alegre’s real 2024 loss far worse than the ”South America commerce” table did. It is not possible to say with confidence, from the data currently available, whether this is because the curve is a poor match for Porto Alegre’s residential building stock specifically, or because that stock is itself atypical of the regional average the curve represents; resolving this would require a depth-based footprint of the 2024 event, the same limitation already noted in Section 2.5.

This is not a computational error. It is a legitimate, if uncomfortable, finding: at the best available estimate of real flood risk, Residential adaptation as currently scoped and costed in this exercise is not economically justified, while Companies’ portfolio remains justified with reduced margins. The practical implication is not that residential adaptation should be abandoned, but that it should be re-scoped: the measures here were specified at city scale, and the analysis suggests the case for them has to be made either at a smaller, better-targeted scale or on grounds other than avoided direct economic damage, such as displacement, health, or continuity of services, none of which this model prices.

5 Discussion and Conclusions

The exercise confirms that extending a cost-benefit example from one asset to others within the same case study is not a mechanical exercise in copying code. Calibrating a new asset requires a search space of its own, sensitive to that asset’s exposure scale, and integrating calibration with cost-benefit analysis requires explicit attention to the physical unit of the hazard used at each step: Residential and Companies are calibrated against a duration hazard for event reconstruction, but the cost-benefit analysis runs on a depth hazard, so it uses a separate, depth-based curve (Section 2.5). A declared physical unit compatible with the hazard actually used is necessary but not sufficient: the literature curve’s amplitude, tested with no adjustment against the real loss observed in 2024, overestimated that loss by 20.7 times for Residential and 2.8 times for Companies, which is why its amplitude is calibrated locally rather than taken at face value. The broader lesson is that a damage curve’s physical unit and its magnitude are two separate properties to verify; a curve can be dimensionally correct and still produce substantially incorrect results, silently, because the output looks plausible enough not to raise suspicion unless it is deliberately checked against a real observed loss.

From a decision-making standpoint, the final picture is more heterogeneous than a single conclusion of the form ”adaptation measures are worthwhile.” For Companies, the six-measure portfolio makes economic sense, with modest margins but all above break-even. For Residential, none of the six parametrized measures pays for itself against the real estimated risk: not the wide-infrastructure measure (drainage), not the partial relocation, not the MDD or PAA measures. This does not mean adaptation is not worthwhile for Porto Alegre’s residential sector; it means the measures *as parametrized in this exercise*, at these costs and this scale, are not jus-

tified against the best available estimate of risk. A future refinement would need to reconsider the scale of the interventions, for example targeting only the highest-risk points rather than the whole city, or seek genuine cost reductions, rather than simply accepting the current figures. Parametric insurance, across all three assets, remains structurally the lowest benefit-to-cost mechanism (approximately 0.66), a cost of transferring volatility worth explaining to a decision maker as expected, not as a poorly parametrized measure.

The most important limitation of this exercise is that the final damage curve, though anchored to the real 2024 loss, was calibrated against a single aggregate data point, the event's total loss, not against a spatial distribution of losses like the one that guided the duration hazard's calibration in Section 2.3. The 200-year return period used as a proxy for 2024's depth was chosen by area similarity between two distinct hazard models, the 2024 duration model and the return-period depth model, not by a direct physical correspondence; it is the best available approximation, not a calibration with the same rigor as Section 2.3. Any future refinement of the Brazil case study should prioritize obtaining or reconstructing a depth-based footprint of the 2024 event, to enable a direct, spatially distributed calibration, the way one already exists for the duration hazard.

6 Recommendations for Replication

Every number in this report is reproducible from the data and code delivered alongside it, without needing to rebuild anything from scratch. The five notebooks that produce the results in Section 3, the calibration in 06b and the four cost-benefit runs in 09 through 12, can each be re-executed independently once the environment described in Section 2.1 is set up, and each writes its own figures and printed totals in place, so a discrepancy in any one number can be traced back to the specific notebook cell that produced it. The commands below assume a terminal open in the root of this repository, with the `climada_env` environment already created and activated.

```
conda activate climada_env
cd notebooks/
jupyter nbconvert --to notebook --execute --inplace \
    06b_calibration_residential_companies.ipynb
jupyter nbconvert --to notebook --execute --inplace 09_costbenefit.ipynb
jupyter nbconvert --to notebook --execute --inplace \
    10_costbenefit_additional_measures.ipynb
jupyter nbconvert --to notebook --execute --inplace \
    11_costbenefit_residential.ipynb
jupyter nbconvert --to notebook --execute --inplace \
    12_costbenefit_companies.ipynb
```

All required input files, including the calibrated and measure-parametrized Entity files and the relocation HDF5 files, are already available in `data/`. It is not necessary to run the numbered `.py` scripts to reproduce the results in this report; they exist only as a record of how each file was built and are documented in `notebooks/README.md`. Running them is only necessary to rebuild the inputs from scratch or to adapt part of the exercise.

To reproduce or extend this analysis within the Brazil case study, five general steps apply. First, calibrate the asset of interest following the method of notebooks 06a and 06b, adjusting the search space to that asset's own exposure scale, since the original search space does not transfer between assets of different scales. Second, and most important, confirm that the physical unit of the hazard used in calibration matches the physical unit of the hazard that will be consumed in the cost-benefit analysis; this should be a mandatory step before any calibration,

since a unit mismatch produces no exceptions and no error messages, only numerically plausible and substantially incorrect results. Third, if a literature curve is substituted in to resolve a unit or domain mismatch, validate its magnitude against the best available local loss data before trusting absolute dollar figures, even a single aggregate observation, as Section 2.5 does; do not assume that a dimensionally correct curve is automatically a calibrated one. Fourth, design adaptation measures to cover distinct CLIMADA mechanisms, since varying the mechanism, not just the numeric value of the parameters, is what makes the analysis informative for the decision maker. Fifth, refine the parametrization against the three acceptance criteria in Section 2.7 before treating the final numbers as final, and resist the temptation to retune costs purely to make an uncomfortable result look better.

References

- [1] CLIMADA development team. *CLIMADA documentation*, 2026. URL <https://climada-python.readthedocs.io/en/v6.1.0/>. Version 6.1.0.
- [2] J. Huizinga, H. De Moel, and W. Szewczyk. Global flood depth-damage functions: Methodology and the database with guidelines. EUR JRC105688, Joint Research Centre (JRC), European Commission, 2017. URL <https://publications.jrc.ec.europa.eu/repository/handle/JRC105688>.

A Technical Notes and Recommendations for the Starter Pack

The following notes were compiled while extending the pack for this exercise, for discussion with the UNU-EHS team and for other participants working on the Brazil case study or extending their own. They cover friction points in the Starter Pack’s data structures, plotting utilities, and calibration workflow that were not obvious from the existing teaching material, but that materially affected how long this exercise took and how easy the results are to trust. Some are narrow, single-line fixes; others, such as the distinction between a dimensionally correct curve and a locally validated one, point to a broader methodological habit worth building into the pack’s documentation rather than left for each participant to rediscover independently. They are ordered roughly by how early in a typical extension of the pack a participant is likely to encounter them.

1. **Exposures-swap measures interact badly with economic growth, and the pack does not warn about it.** The `assets` file measure column points to a single HDF5 file, but `CostBenefit.calc` applies the present Entity’s measures to the present scenario and the future Entity’s measures to the future scenario. If both Entities’ relocation measure point at the same, present-values HDF5 file, the measure’s benefit silently includes the removal of decades of economic growth from the adapted future scenario, overstating it substantially. This exercise handled it by writing a second, growth-scaled HDF5 file and repointing the future entity’s measure in code. A short warning in the cost-benefit notebook, or a worked example, would save other participants from this trap.
2. **No worked example derives insurance parameters from the model itself.** The training material introduces `Impact.calc_freq_curve()` and the risk transfer attachment and cover parameters separately, but does not connect them. This exercise set attachment and cover from the present-day unadapted exceedance curve, which felt like the natural pattern to teach; it is worth adding to the notebook, since arbitrary first-guess constants produce very different premiums and benefits.

3. **Relative paths in the `assets` file column resolve against the process working directory, not the Entity file's own location.** This works while notebooks run from the `notebooks/` folder, but breaks the moment an Entity file is loaded from anywhere else. This friction recurred twice during this session's own corrections: once when copying a corrected Entity file between two differently named folder layouts, and again when renaming those same folders for public release, both times breaking the relocation measure silently until re-executed and checked. Resolving `assets` file relative to the Entity file's own location, rather than the process working directory, would remove this entire class of bug.
4. **A dimensionally correct curve is not the same as a validated curve.** The pack is careful about calibration uncertainty where it calibrates explicitly, but the depth-based curve shipped with the Schools entity arrives pre-made, with no note of its provenance or of whether its amplitude was ever checked against local data. Adopting the equivalent literature curve for Residential and Companies and then checking it against Porto Alegre's real 2024 loss (Section 2.5) showed the unscaled version overestimating that loss by roughly 20.7 times for Residential and 2.8 times for Companies. Labelling the shipped curve with its source and adding a single sentence next to any literature-curve suggestion, stating that it is a regional default whose amplitude must be validated against available local loss data before absolute figures are trusted, would help other participants catch this earlier.
5. **The expert-judgement bounds that resolve the calibration's degeneracy are asset-specific, and nothing warns you when they stop applying.** The pack handles this well: 06a explains clearly that fitting two parameters to one observation leaves a whole family of equally good solutions, and resolves it with stated expert judgement for Schools, `max_damage > 0.7` because several schools were rebuilt outright, and `max_intensity < 30` read off the parameter-space chart as the consequence of that first choice. The difficulty is that the second bound is derived from the first, and only the second survives transfer to another asset: Residential and Companies calibrate to damage ratios of 0.028 and 0.294, so `max_damage > 0.7` is impossible by construction, while `max_intensity < 30` still runs and still silently selects a result. A participant adapting 06a to a new asset therefore inherits a numerical bound stripped of its justification. Restating the expert-judgement cell as a short checklist, asking which bounds are asset-specific and what evidence would set them for the new asset, would make that dependency visible at the point where it matters.
6. **`plot_basemap_eai_exposure()`'s automatic colorbar collides with a custom title once values exceed 10^6 .** Matplotlib adds a scientific-notation offset label above the colorbar, at the same height as a centered axis title; for Schools, values under 200,000, this never appears, but Residential and Companies, values up to several million USD, triggered it. The fix is to grab the colorbar axis and force a plain `ScalarFormatter` instead of fighting it with title length or position. Also, calling `ax.set_title(..., loc="left")` after a function that already sets its own centered title creates a second, independent title slot rather than replacing the first, since matplotlib stores left, center, and right titles separately; call `set_title` without a `loc` argument to replace it.
7. **Windows-specific kernel failure, not reproduced on Linux.** On Windows, the conda-forge `pyzmq/libzmq` build can make every `jupyter nbconvert --execute` fail with `Bad file descriptor (epoll.cpp:73)`; replacing `pyzmq` with the PyPI wheel (`pip install --force-reinstall pyzmq==26.2.0`) restores kernel communication. This exercise's own notebooks were re-executed and verified on Linux without encountering this issue, so it is specific to the Windows build of the conda-forge package, not to the notebooks themselves; it is still worth a line in the installation FAQ for participants who work

on Windows.

8. **Insurance's benefit-to-cost ratio is approximately one over the cost factor by construction, and that is worth stating explicitly.** With a premium loading of 1.5, the ratio lands near 0.66 regardless of the layer. The measure's value is transferring volatility, protection against rare, budget-breaking years, not expected-value savings; a sentence in the teaching material would preempt an "insurance looks bad" misreading of the bar chart.
9. **06a's Excel export step is manual and mislabels the intensity unit.** The final cell of `06a_calibration_schools.ipynb` prints the calibrated functions with column `intensity_m` and `intensity_unit: "meter"`, even though the calibration ran on flood duration in days, and the table must then be pasted into the calibrated workbook by hand. Notebook **06b** automated this export with `openpyxl` and wrote the unit as `days`. It is also worth clarifying that the *shipped* calibrated Schools entity uses a depth-based curve consistent with the JRC return-period rasters, which is not the duration-based curve **06a** itself produces; the connection between the two would be good to explain explicitly in the teaching material.